

University of Dhaka



Institute of Information Technology

Course Name : Computer Networking
Course code : CSE 2111
Assignment : Wireshark Lab

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1. HTTP version 1.1
2. Accepted Language: en-US
3. IP: 10.100.202.69
4. Status Code: 200
5. Last Modified: Tue, 28 Oct 2025 05:59:01 GMT
6. Size: 128 bytes

```
⌵ HTTP/1.1 200 OK\r\n
  Response Version: HTTP/1.1
  Status Code: 200
  [Status Code Description: OK]
  Response Phrase: OK
  Date: Sun, 01 Feb 2026 09:07:14 GMT\r\n
  Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.1 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
  Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
  ETag: "80-64231b6717e87"\r\n
  Accept-Ranges: bytes\r\n
> Content-Length: 128\r\n
  Keep-Alive: timeout=5, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=UTF-8\r\n
\r\n
\[Request in frame: 25\]
[Time since request: 305.264000 milliseconds]
[Request URI: /wireshark-labs/HTTP-wireshark-file1.html]
\[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html\]
File Data: 128 bytes
```

HTTP response

	Time	Source	Destination	Protocol	Length	Info
25	6.351447	10.100.202.69	128.119.245.12	HTTP	490	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
28	6.656711	128.119.245.12	10.100.202.69	HTTP	555	HTTP/1.1 200 OK (text/html)

List of packet captured

```
⌵ Hypertext Transfer Protocol
  ⌵ GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /wireshark-labs/HTTP-wireshark-file1.html
    Request Version: HTTP/1.1
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/26.2 Safari/605.1.15\r\n
    Upgrade-Insecure-Requests: 1\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    Priority: u=0, i\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
\r\n
\[Response in frame: 28\]
\[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html\]
```

HTTP request

8. No

9. Yes, Browser returns a file as the status code: 200 OK.

```
✓ HTTP/1.1 200 OK\r\n
  Response Version: HTTP/1.1
  Status Code: 200
  [Status Code Description: OK]
  Response Phrase: OK
```

HTTP response

10. Yes, If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT

```
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9,bn;q=0.8\r\n
If-None-Match: "173-64231b67176b7"\r\n
If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT\r\n
```

Language

11. Status code: 304 Not Modified

```
✓ HTTP/1.1 304 Not Modified\r\n
  Response Version: HTTP/1.1
  Status Code: 304
  [Status Code Description: Not Modified]
  Response Phrase: Not Modified
```

HTTP response

12. 1 GET request. The first HTTP packet in the filtered trace
13. The first packet.
14. Status code: 200 OK
15. 4

```

v HTTP/1.1 200 OK\r\n
  Response Version: HTTP/1.1
  Status Code: 200
  [Status Code Description: OK]
  Response Phrase: OK

```

HTTP response

16. 3 http get requests. Destination ip: 128.119.245.12

Time	Source	Destination	Protocol	Length	Info
108	11.266713	192.168.0.102	128.119.245.12	HTTP	553 GET /wireshark-labs/HTTP-wi
118	11.577406	128.119.245.12	192.168.0.102	HTTP	112 HTTP/1.1 200 OK (text/html
123	11.614259	192.168.0.102	128.119.245.12	HTTP	499 GET /pearson.png HTTP/1.1
126	11.924667	128.119.245.12	192.168.0.102	HTTP	648 HTTP/1.1 301 Moved Permanen
176	12.472646	192.168.0.102	2.56.99.24	HTTP	466 GET /8E_cover_small.jpg HTT
810	14.205757	2.56.99.24	192.168.0.102	HTTP	1230 HTTP/1.1 200 OK (JPEG JFIF

17. The two images were downloaded in parallel. This is evident because the browser sends HTTP GET requests for both images within a very short time interval and maintains multiple simultaneous TCP connections to different servers. The overlapping timestamps and concurrent TCP streams in Wireshark indicate that the browser did not wait for one image to be fully downloaded before requesting the other.

```

v Hypertext Transfer Protocol
  > HTTP/1.1 401 Unauthorized\r\n
    Date: Thu, 05 Feb 2026 03:01:55 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.1 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n

```

18. 401 Unauthorized
19. Yes, this is the second HTTP GET request. It includes the Authorization header containing the Base64-encoded username and password, which was added after the server responded with a 401 Unauthorized message to the initial GET request.

```

v Hypertext Transfer Protocol
  v GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /wireshark-labs/protected_pages/HTTP-wireshark-file5.html
    Request Version: HTTP/1.1
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
  > Authorization: Basic d2lyZXNoYXJrLXN0dWRlbmRzOm5ldHdvcm==\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) C
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9,bn;q=0.8\r\n
    \r\n

```